

Probability of default, done point-in-time

A consumer credit underwriting model on 779,025 Lending Club loans, 2007 to 2016 vintages, and the lending policy that follows from it.

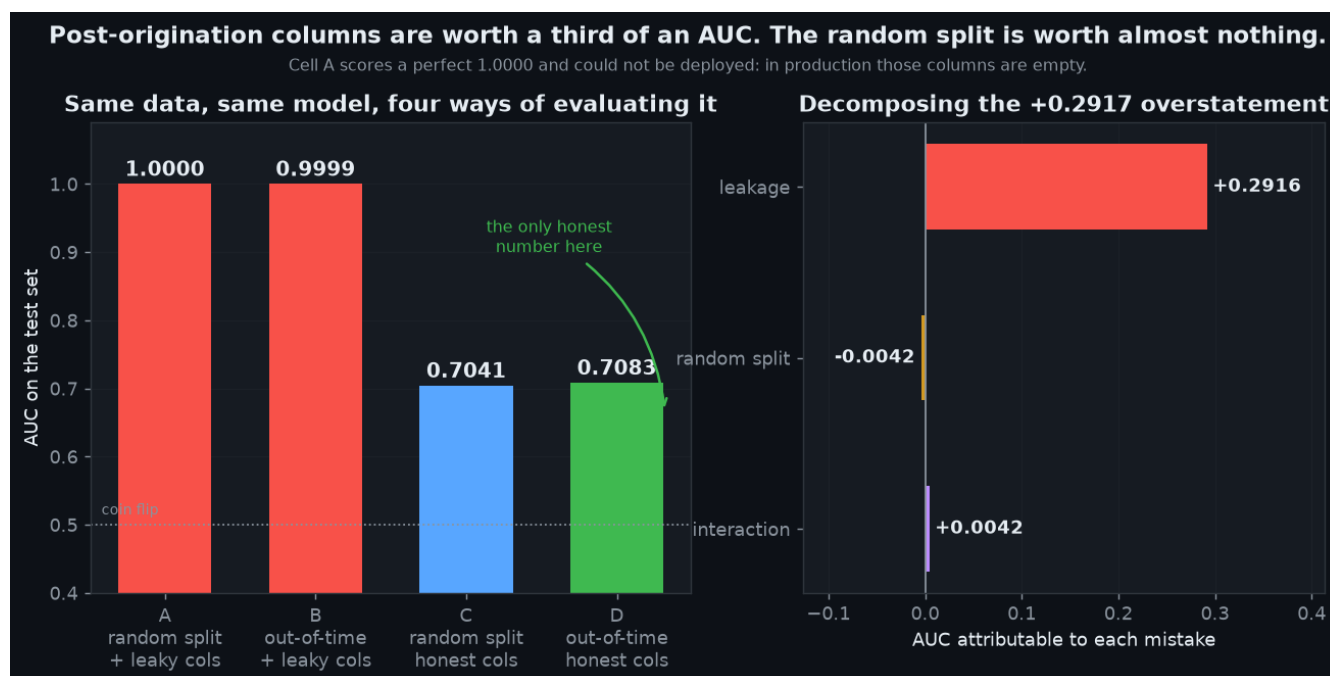
Most published loan-default models report an AUC somewhere between 0.95 and 0.99. None of them could be deployed. The reason is not subtle statistics, it is data lineage: the dataset contains dozens of columns recorded *after* the loan was issued, and at the moment an underwriter has to decide, every one of them is empty. This project measures exactly what that mistake is worth, builds the model that does not make it, and then turns it into an approve/decline policy using arithmetic on money rather than a threshold of 0.5.

What the two classic mistakes are worth

Configuration	Split	Columns	AUC
A careless	random	all, including post-origination	1.0000
B	out-of-time	all, including post-origination	0.9999
C	random	origination only	0.7041
D honest	out-of-time	origination only	0.7083

Cell A is a perfect classifier — 1.0000 — and is worth nothing. Holding one mistake fixed at a time separates them: post-origination columns account for **+0.2916** of the +0.2917 gap. The random split accounts for **-0.0042**.

That second figure is the opposite of what I predicted before running it, and it is reported as a failed hypothesis rather than quietly dropped. The mechanism is the population rule described below: once loans are required to have had their full term to mature, the 2017 and 2018 vintages disappear entirely and every remaining year defaults at 14 to 16 percent. There is almost no temporal drift left for a random split to exploit. On a dataset without that filter, the same test would very likely show the textbook result.



The worst offender is not the obviously named one. `last_fico_range_high` scores 0.9173 entirely on its own — better than recoveries at 0.8940 — because it is the borrower's credit score at the most recent bureau pull, which on a defaulted loan happened after the default wrecked it. Sorted alphabetically it sits next to the legitimate `fico_range_high`, which is precisely how it ends up in a feature list by accident.

The population, and why it is smaller than it looks

Two rules, both non-negotiable. A loan needs a resolved outcome, which removes everything still in flight. And it needs to have had its full contractual term before the snapshot — a 60-month loan issued in 2017 cannot appear as Fully Paid in a 2018 Q4 extract. Filtering on resolved status alone keeps recent vintages *only if they defaulted early*, which is survivorship bias pointing the wrong way.

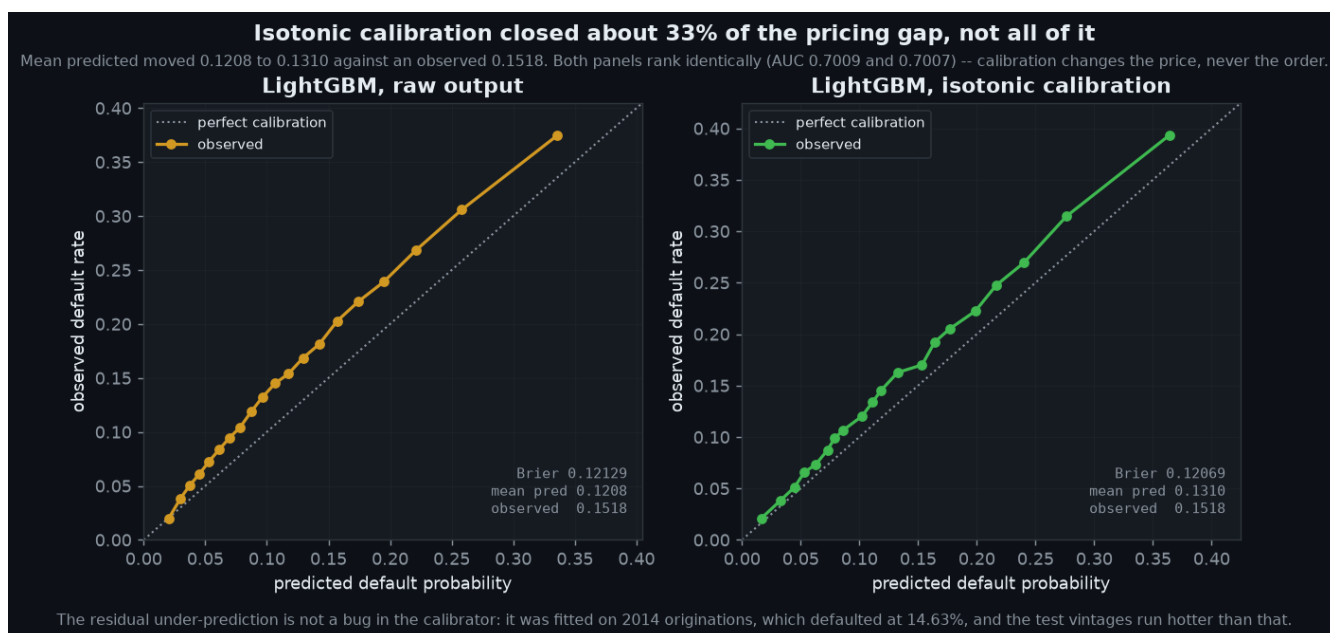
Stage	Loans	Removed
All loans in the extract	2,260,701	
Resolved outcome only	1,348,059	912,642 still in flight
Full term before snapshot	779,025	569,034 not yet matured
Charged off	118,371	15.19% default rate

The maturity rule keeps 100% of 2013 originations, 30.6% of 2016, and **none at all** of 2017 and 2018. Losing a quarter of a million recent loans is painful and it is still correct.

Models

Model	AUC	Gini	KS	Brier
Constant baseline	0.5000	0.0000	—	0.12877
WOE scorecard (27 variables)	0.6892	0.3783	0.2752	0.12146
LightGBM	0.7009	0.4019	0.2919	0.12129
LightGBM, isotonic calibrated	0.7007	0.4014	0.2918	0.12069

Accuracy is deliberately absent. At a 15% default rate, predicting that everybody repays scores 85% and is useless. Calibration improves Brier from 0.12129 to 0.12069 while leaving AUC unchanged at 0.7007 — calibration changes the price, never the order.



From probability to policy

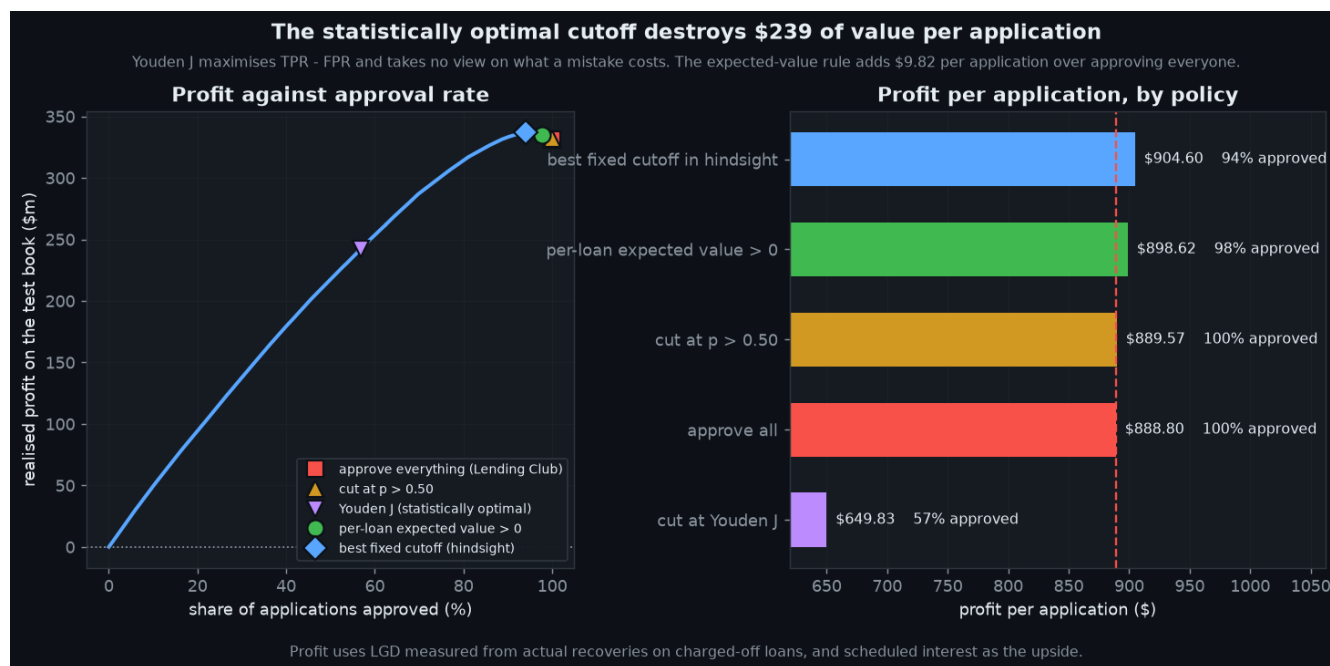
A probability is not a decision. Approving a loan is worth $(1-p) \cdot \text{interest} - p \cdot \text{LGD} \cdot \text{exposure}$, so it is worth doing while p sits below $\text{interest} / (\text{interest} + \text{LGD} \cdot \text{exposure})$. That break-even is a property of the individual loan, not a global constant: across this test book the middle 90% of loans break even between 14.9% and 36.2%, median 24.9%, because a 60-month loan at 26% can carry far more risk than a 36-month loan at 7%. Loss given default is measured from actual recoveries on loans that charged off — mean 0.539, median 0.562.

16 of the 372,810 loans (0.004%) have scheduled payments that do not exceed principal, so they earn no interest and break even at zero. They can never clear an expected-value test, which is the correct outcome for a loan with no upside.

Policy	Approved	Book bad rate	Profit / application
approve all (what LC did)	100.0%	15.18%	\$888.80
cut at $p > 0.50$	100.0%	15.17%	\$889.57
cut at Youden J (0.1218)	56.8%	8.57%	\$649.83
per-loan expected value > 0	97.6%	14.70%	\$898.62
best fixed cutoff in hindsight (0.2800)	94.0%	13.80%	\$904.60

The statistically optimal cutoff is a disaster. Youden's J maximises TPR – FPR, declines 43% of applications, and destroys \$239 of value per application against simply approving everyone. The expected-value rule adds \$9.82 per application instead — about 1.1% on a \$331m book.

That 1.1% deserves context rather than spin. Every loan in this dataset was already approved by Lending Club's own underwriting, so the model is finding residual risk among applicants who already passed a credit screen. The easy declines happened upstream and are not in the data. A model sitting at the front of the funnel would have far more room; this number is the honest one for the population available.



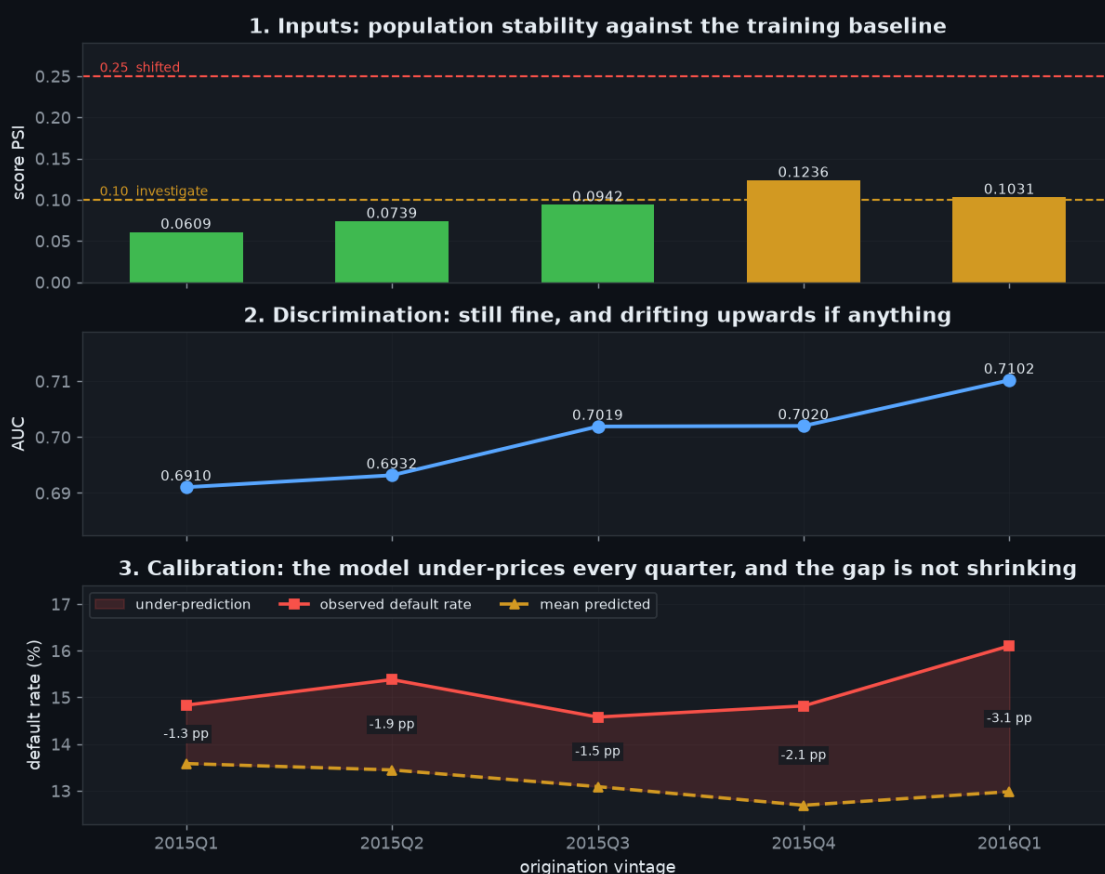
Monitoring, and the price of interpretability

Three views of the same five quarters, and they disagree — which is the whole argument for having all three. Score PSI climbs from 0.0609 to 0.1031 and crosses the conventional 0.10 line. Discrimination does not degrade at all; AUC actually improves from 0.6910 to 0.7102. But the model under-predicts the default rate in every single quarter, and the gap widens from 1.3 to 3.1 percentage points. Any one signal alone gives the wrong answer.

Most-shifted variable	PSI	Verdict
initial_list_status	0.4904	shifted
mths_since_last_record	0.3225	shifted
int_rate	0.2977	shifted
sub_grade	0.0964	stable

The largest shift is `initial_list_status`, which records whether Lending Club listed a loan whole or fractionally. That is a change in their own platform operations, not in borrower quality — exactly the distinction per-variable PSI exists to make. Reacting to it by retraining would be solving the wrong problem.

PSI watches the inputs, AUC watches the ranking, calibration watches the price. All three, or none.



Isotonic calibration was fitted on 2014 originations, which defaulted at 14.63%. The test vintages run hotter, so the model under-predicts even after calibrating.

The promotion gate

The interpretable scorecard was run as a challenger against the LightGBM champion, and it **failed** on discrimination, calibration and profit. It failed narrowly: 0.6892 AUC against 0.7007, and \$332.1m profit against \$335.0m on a \$335m book. So the cost of being able to explain a decline is roughly 0.012 AUC and 0.86% of profit. That is a number a credit committee can accept or reject, which is more useful than an opinion about interpretability.

Gate decision: **KEEP CHAMPION**. Failing is the expected outcome most of the time; that is the point of having a gate rather than a preference.

What I would do next

The residual under-prediction is the real open problem. Isotonic calibration was fitted on 2014 originations, which defaulted at 14.63%, and the test vintages run hotter — so the calibrator is faithfully reproducing a base rate that has since moved. A rolling recalibration on the most recent closed book, rather than a fixed slice, is the obvious fix and the monitoring already detects the need for it. Beyond that: reject inference, since this population is already screened and the declined applications are invisible; and survival modelling of time-to-default, which would let the maturity rule be relaxed instead of discarding a quarter of a million loans.

Data: Lending Club accepted loans, 2007–2018 Q4, 2,260,701 rows, 151 columns, SHA-256 verified at download. All figures generated by the pipeline in this repository; nothing in this document is typed by hand.